



Applied Artificial Intelligence

Module Code	Workload	Credits/LP	Semester	Frequency of Module	Duration
	180 h	6	6	Each semester	1
1	Module Components Applied Artificial Intelligence	Teaching Language englisch	Contact Hours 45 h	Self Study 135 h	Class Size 20
2	Learning outcomes / Competencies After successfully completing the module, students will be able to... Knowledge (1) - describe the use of Artificial Intelligence in business - Outline procedures for a successful Data Science project. Understand (2) - Understand concepts of data sourcing and data analysis - Explain how different machine learning algorithms work Application (3) - Design a Data Science project with business application - Present the implementation from data sourcing to business application Analysis (4) - Evaluate Data Science Project Synthesis (5) - Implementation of a Data Science project				
3	Contents Machine learning is a method of data science that automates analytical model building. It is a branch of artificial intelligence based on the idea that systems can learn from data, identify patterns and make autonomous decisions. The course deals with the study of computer algorithms that improve automatically (themselves) through experience. • Data - Data acquisition - Data analysis - Data visualization • Modeling - Business idea - Data preparation - Prediction - Evaluation • Implementation of own Data Science project - Presentation of the project - Challenging of another project				

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.0	spet	01.06.2021/spet	01.09.2021



4	Teaching Methods Workshop
5	Prerequisites - Databases - Applied Mathematics - Programming Fundamentals
6	Methods of Assessment presentation, delivery, discussion of other project(s) (40% / 40% / 20%)
7	Applicability of Module Elective module in the bachelor's degree program International Business Information Systems (IBS)
8	Person Responsible for Module Prof. Dr. Peter Schanbacher
9	Reading List - Alpaydin, E.: Introduction to Machine Learning, MIT Press, 2010 - Witten, E. et al.: Data Mining: Practical Machine Learning Tools and Techniques, Morgan Kaufmann, 2011 - Janert, Philipp K. Data analysis with open source tools. " O'Reilly Media, Inc.", 2010. - Provost, Foster, and Tom Fawcett. Data Science for Business: What you need to know about data mining and data-analytic thinking. " O'Reilly Media, Inc.", 2013. - Marz, Nathan, and James Warren. Big Data: Principles and best practices of scalable realtime data systems. Manning Publications Co., 2015. - Grus, Joel. Data Science from Scratch. First principles with Python. O'Reilly. 2017. - O'Neil, Catherine; Schutt, Rachel. Doing data science. 2nd release. O'Reilly. 2013.

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.0	spet	01.06.2021/spet	01.09.2021

Performance Marketing & Marketing Automation applied on E-Commerce					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	4, 5, 6	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Performance Marketing	a) English	a) 45 hrs.	a) 135 hrs.	a) 15
2	Performance Marketing refers to online marketing and advertising programs in which advertisers and marketing companies are paid when a specific action/result is completed/achieved; such as a sale, lead or click. The performance marketing program includes analysis of the given marketing and sales situation, planning of campaigns, choice of appropriate measures, selection and application of tools and evaluation of results and ROI of measures to optimize the performance of campaigns and achieve the defined goals. Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... describe the essential measures for search engine optimization (SEO). ... know in which steps to create a Performance Marketing Campaign ... know how to measure and optimize the performance of an email campaign ... give an overview of methods for marketing automation ... quantify performance benchmarks in terms of conversion rate in various industries ... list different methods for targeting and attribution ... list different approaches of Search Engine Optimization (SEO) ... list top 10 tactical measures of SEO ... recite the different goals, advantages and disadvantages of SEO, SEA, Display marketing and Affiliate marketing Comprehension (2) ... explain the different purposes and roles of a Data Management Platform ... explain a typical conversion funnel and the meaning for lead management ... explain different pricing models of Performance Marketing Platforms and typical pricing of PM Channels ... explain different measures to optimize the performance of a website and landing page ... explain the basic functions and name examples of a Demand Side Platform and a Supply Side Platform ... explain the function of a Media Optimization Platform Application (3) ... apply tools in order to optimize the copy of editorial content for search engines ... apply tactics to drive qualified leads through a website ... plan Google Advertising campaigns ... identify, compare and evaluate tools for marketing automation ... analyse key performance indicators to evaluate results of campaigns ... decide on which factors to focus to optimize the ranking of a website in search engines Analysis (4) ... analyse key words used by consumers in search engines to identify market potentials of specific demand ... evaluate the pros and cons and appropriate application cases of different attribution models in performance marketing ... evaluate appropriate keywords and explain the advantages of long tail keywords				

3	Individual Component Content - Introduction to Performance Marketing (goals, data driven advertising, channels, data management platform) - Strategic Planning in Performance Marketing (planning, measurement, optimization, digital marketing strategy) - Evolution of Instruments for performance marketing (SEO, SEA, Social Media, Display, Affiliate, Email) - SEO Search Engine Optimization (unpaid traffic through optimized content) - <i>Case Study & Application Project</i> - SEA Search Engine Marketing (scalable traffic with paid advertising) <i>Case Study & Application Project</i> - E-Mail Marketing (direct dialogue with users, marketing automation) - Marketing Automation (purposes, process design, tools, methods) - Display Advertising (native, programmatic, DSPs, SSPs, DMP, targeting, retargeting) - Affiliate Marketing (product search, distributions networks, Amazon marketplace) <i>Case Study & Application Project</i> - Social Media Advertising (interest based targeting, social networks, targeting options, ad forms) - Influencer Marketing (multipliers and brand ambassadors) - Web-Analytics (customer journey, attribution, KPIs, metrics for performance measurement)
---	---

4	Teaching Methods a) Seminar with application cases
5	Prerequisites IBS 1: Scientific Working & Academic Writing IBS 2: Digital Marketing & E-Commerce IBS 3: Applied Statistics
6	Methods of Assessment a) Graded Assessment 1A PL (Teaching event in seminar modus, Practical Workshop – several tasks during the lecture period of a semester will be worked out by groups of 3 students each at given various application cases with focus on e-commerce, oral exam during final presentation of results) (6 Credit Points)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Pavel Rawe (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) Kamps, Ingo, et al (2020): Performance Marketing – Einführung in Instrumente, Methoden, Technik Kreutzner, Ralf T. (2020): Praxisorientiertes Online-Marketing – Konzepte, Instrumente, Checklisten Lammenett, Erwin (2020): Praxiswissen Online-Marketing – Affiliates, Influencer, Amazon, SEA, SEO Kingsnorth, Simon (2020): Digital Marketing Strategy Freundt, Tjark (2020): Marketing Performance – How Marketers drive profitable Growth Huey, Craig (2020): The new multichannel, Integrated Marketing: 29 Trends McDonald, Jason (2021): SEO Fitness Workbook – Seven Steps to SEO Coombe, Will (2020): 3 Months to No. 1 – SEO Playbook for getting your website found on Google Zehender, Randal (2020): The Basics of Programmatic Advertising – Strategies to increase Performance and hit CPA-Targets Marshall, Perry (2020): Ultimate Guide to Google Ads Geddes, Brian (2020): Advanced Google Adwords Ulander, Kevin (2020): Affiliate Marketing – Beginner's step by step guide Jeffery, Mark (2020): Data-Driven Marketing – The 15 Metrics in Marketing Sorgor, Stephan (2020): Marketing Analytics – Strategic Models and Metrics Tonkin, Sebastian (2020): Performance Marketing with Google Analytics: Strategies – Strategies and

Tactics to maximize online ROI

Grdodian, Greg (2020): Marketing AI – From Automation to Revenue Performance Marketing

Web-Programming

Module code	Workload	Credits/CP	Semester	Frequency of module		Duration
	180 h	6	1			1 Semester
1	Module Web-Programming		Teaching Language English	Contact hours 4 SWS / 60 h	Self-study 120 h	Class size 30
2	Learning outcomes After passing this module successfully, students are able to ... Knowledge (1) • Programming Applications with Web-Technologies • Architectures and Frameworks Understanding (2) • Distributed Applications in WWW Practice (3) • Programming on client site (i.e. JavaScript) • Programming Frameworks (i.e. jQuery / JEE) Analysis (4) • analyse programming solutions • analyse software design					
3	Individual component content • Internet and Web-Technologies • Communication over IP Client-Server • Software and Framework Architectures • Communication over IP • Scripting-Languages • Programming-Languages					
4	Teaching methods • Lectures style • Exercises with computer					
5	Prerequisites • Basic principles in programming					
6	Methods of assessment • Final written exam • Presentation of programs on computer					

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.0	SN	x	x

7	Applicability of module Mandatory
8	Person responsible for module/ lecturer Prof. Dr. Stefan Noll

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.0	SN	x	x

Reading list

- JavaScript: The Definitive Guide, 6th Edition, O'Reilly Media, David Flanagan, 2011
- jQuery Pocket Reference, O'Reilly Media, David Flanagan, 2010
- Head First Servlets and JSP, O'Reilly Media, Bryan Basham et al., 2008
- Head First PHP & MySQL, O'Reilly Media, Lynn Beighley et al., 2008

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.0	SN	x	x

Business Process Automation using Hyperautomation Tools (RPA)

Kennnummer		Workload	Credits/CP	Semester	Häufigkeit des Angebots	Dauer
		180 h	6	1	Each semester	1 Semester
1	Lehrveranstaltungen		Sprache	Kontaktzeit	Selbststudium	Geplante Gruppengröße
	a) Process Optimization based on Process Mining and RPA		a) English	4 SWS / 45 h	135 h	a) 15
	b) Process Optimization based on Process Mining and RPA		b) English			b) 15
2	Lernergebnisse					
	Automating core business processes is critical for organizations looking to succeed in today's highly competitive global market. Companies that want to build and grow need automation to squeeze more performance and business insights out of operations that historically have been managed by people. However organizations often lack the tools to understand and optimize their processes. Besides providing a general overview on hyperautomation technologies this lecture will focus on the following 3 hyperautomation technologies: • Process Mining – Data Mining for Processes and RPA • RPA– Robotic Process Automation • Conversational UI / Chatbots By the end of the lecture, students learn to get an entry in the world of business process optimization using hyperautomation tools by • understanding the benefits and describing the application areas of process mining • applying process mining techniques on a real dataset - getting hands-on experience using Celonis process mining tool • understanding the benefits and describing the application areas of robotic process automation • applying robotic process automation - getting hands-on experience using UiPath RPA tool • understanding the benefits and describing the application areas of Chatbots • developing Chatbots using SAP conversational AI					
3	Inhalte					
	1. Overview Hyperautomation Tools 2. Process Mining – Data Mining for Processes – Foundation of Process Mining – Process Mining Deep Dive with hands-on experience with Celonis toolset 3. RPA – Robotic Process Automation Foundation – Foundation of RPA – RPA Deep Dive with hands-on experience with UiPath toolset					

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.3	jr	QM-Board 11.4.2012, 16.01.2013 04.06.2013/jr	04.06.2013

	4. Chatbots – Chatbots Foundation – Chatbot Deep Dive with hands-on experience with SAP Conversational AI
4	Lehrformen a) Praktikum/Labor b) Vorlesung
5	Teilnahmevoraussetzungen • Basic understanding of Business Process Modelling
6	Prüfungsformen a) Studienleistung 1sbA (Praktische Arbeit) (2 LP) b) Prüfungsleistung 1K (Klausur) (4 LP)
7	Verwendung des Moduls •
8	Modulbeauftragte/r und hauptamtliche Lehrende Prof. Dr. Jürgen Weiner (Module Responsible)
9	Literatur a) Will van der Aalst: Process Mining – Data Science in Action, Second Edition, Springer b) Gartner: Market Guide for Process Mining - Published 17 June 2019 - Analysts Marc Kerremans c) Gartner: Top 10 Strategic Technology Trends for 2020 https://www.gartner.com/en/doc/432920-top-10-strategic-technology-trends-for-2020 d) McKinsey: Four Fundamentals of Workplace Automation. https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/four-fundamentals-ofworkplace-automation e) Craig Le Clair: The Forrester Wave™: Robotic Process Automation, Q4 2019 f) Tom Taulli: The Robotic Process Automation Handbook: A Guide to Implementing RPA Systems, Apress g) Alok Mani Tripathi: Learning Robotic Process Automation: Create Software robots and automate business processes with the leading RPA tool – UiPath (English Edition) (Englisch) h) Gartner: Gartner Predicts 25 Percent of Digital Workers Will Use Virtual Employee Assistants Daily by 2021 https://www.gartner.com/en/newsroom/press-releases/2019-01-09-gartner-predicts-25-percent-of-digital-workers-will-u i) Deloitte Digital: Conversational AI – The next wave of customer and employee experiences: https://www2.deloitte.com/content/dam/Deloitte/au/Documents/strategy/au-deloitte-conversational-ai.pdf

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.3	jr	QM-Board 11.4.2012, 16.01.2013 04.06.2013/jr	04.06.2013

AUSFÜLLHILFE: BEWEGEN SIE DEN MAUSZEIGER ÜBER DIE ÜBERSCHRIFTEN. AUSFÜHRICHE HINWEISE: [LEITFADEN MODULBESCHREIBUNG](#)

SAP Business Configuration and ABAP Programming						
Module Code		Workload	Credits	Semester	Frequency of Module	Duration
		180 hrs.	6	1	Summer (on request)	1 Semester
1	Module Components		Teaching Language	Contact Hours	Self Study	Class Size
	a) SAP Business Configuration und ABAP Basics - Exercises		a) English	45 hrs.	135 hrs.	a) 30
	b) SAP Business Configuration and ABAP Basics/Fundamentals,		b) English			b) 30
2	Learning Outcomes					
	Configuration and Customizing are important activities implementing an ERP-System.					
	Configuration					
	ERP vendors embed multitude of configurable options in their solutions that companies can select in order to tailor the software specifically to their needs. You will learn to configure important business processes like Order-to-Cash, Procure-to-Pay, Financial and Management Accounting based on the ERP system S/4HANA from SAP.					
	Customization					
	There are circumstances where just configuring the ERP system is not enough, and customizing the software is necessary. Customization consists of altering existing program code or adding program code to make the ERP system fit a company's unique requirement. We will use ABAP (Advanced Business Application Programming) for development and customization purposes in the SAP software. This part of the lectures explains the key concepts of SAP ABAP. By the end of the lecture, you will learn to get an entry in the world of ABAP-programing by					
	• understanding the basic APAP syntax • developing applications starting with basic reports till complex modular dialog programs • creating and maintaining database tables based on the ABAP Dictionary • learning the difference between the procedural and object-orientated ABAP-programming model based on sample application for student-/lecture administration					
3	Content					
	1. Overview: Activities necessary to implement an ERP-System					
	2. Business Configuration based on S/4HANA system					
	– Overview configuration steps – Configuration of Enterprise structure – Configuration of Financial and Management Accounting – Configuration of Procure-to-Pay process – Configuration of Order-to-Cash process					
	2. Customization based on ABAP - Advanced Business Application Programming					
	– Introduction – My First ABAP Program – Development of Basic Reports – Creating Data Dictionary Objects					

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.3	jr	QM-Board 11.4.2012, 16.01.2013 04.06.2013/jr	04.06.2013

	– Data Manipulation – Accessing the database – Dialog Programming with ABAP Dynpro – Object Oriented Programming with ABAP Objects – Enhancements - Lock Concept and Exception Handling
4	Teaching Methods a) Practical Exercise/Laboratory b) Vorlesung/Labor
5	Prerequisites ERP Basic knowledge of other programming languages like Java
6	Methods of Assessment a) Submitted Assignment of Practical Exercise (2 Credits) b) Supervised Exam (4 Credits)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person responsible for Module Prof. Jürgen Weiner (Module Responsible)
9	Reading List (core texts and recommended texts) a) Marianne Bradford: Modern ERP – Select, Implement, and Use Today's Advanced Business Systems, Raleigh b) Brian O'Neill: Getting started with ABAP; Rheinwerk Publishing c) James Wood, Joseph Rupert, Object-Oriented Programming with ABAP Objects, Rheinwerk Publishing

Version	Erstellt von	Freigabe (Datum/Kürzel)	Gültig ab
1.3	jr	QM-Board 11.4.2012, 16.01.2013 04.06.2013/jr	04.06.2013

Digital Risk Management					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	90 hrs.	3	?	BLOCK	1 Semester
1	Module Components		Teaching Language	Contact Hours	Self Study
	a) Risk Management		a) English	a) 21 hrs.	a) 69 hrs.
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... describe the corporate Risk Management/Risk assessment process. ... explain basic Risk Management techniques using financial/physical traded instruments. ... understand the basic principles of financial and energy commodity markets. Comprehension (2) ... explain risk profiles of various financial instruments. ... explain the corporate three lines of defence model and the role of risk management. ... explain most common risk measures and their use in the corporate environment. Application (3) ... elaborate a position view of various contract structures and evaluate such positions. ... calculate risk measures for such positions. ... apply basic risk management concepts to positions using financial instruments (energy market examples). Analysis (4) ... analyse risk positions resulting from contractual agreements/physical assets and devise risk reducing hedging strategies.				
3	Individual Component Content a) Markets and Regulatory Requirements 1. Financial and Energy Markets Introduction 2. Risk Management in Organisations 3. Regulatory Requirements for Banks and Corporates b) Financial Instruments and Positions 1. Traded Instruments (Exchange/OTC) 2. Contractual Positions / Natural Positions 3. Portfolio construction c) Position Valuation and Risk Measures 1. Mark-to-market vs. mark-to-model 2. Value-at-Risk – Concept and Calculation Methods 3. Other Portfolio Risk Measures d) Applied Risk Management 1. Risk Management Process 2. Exposure Hedging 3. Limit Systems and Monitoring e) Digital Developments 1. Risk Management Automation 2. Machine Learning applications for Risk Management				

4	Teaching Methods a) Seminar, one-week block with 14 lectures spread over five days (Mon-Fri)
5	Prerequisites None
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (3 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Dipl.-Kfm. Marcus Craul (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Jovanovic: Hedging Commodities, Harriman House, 2014 b) Da Costa Lewis: Market Risk Modelling, Risk Waters, 2003 c) Jorion: Value at Risk, McGraw Hill, 1997 d) Eydeland, Wolyniec: Energy and Power Risk Management, Wiley, 2003 e) Jackson, Staunton: Advanced modelling in Finance using Excel and VBA, Wiley, 2002 f) Clewlow, Strickland: Implementing Derivative Models, Wiley, 1998 g) Dalton: Financial applications using Excel add-in development in C/C++

Business Intelligence and Data Analytics

Module Code	Workload	Credits	Semester	Frequency of Module	Dauer	
	180 h	6	5 (IBS)	Each semester	1 Semester	
1	Module Components		Language	Contact Hours	Self Study	Class Size
	Business Intelligence and Data Analytics		English	45 h	135 h	16
2	Learning Outcomes After successful participation in the module the students are able to ... Knowledge (1) ... define concepts of business intelligence and data analytics ... list business intelligence and data analytics tools ... name state-of-the-art topics and market trends Comprehension (2) ... explain why investing and using business intelligence and data analytics is crucial for business success ... discuss differences between business intelligence approaches and data analytics projects ... illustrate business intelligence and data analytics technical architecture ... recognize machine learning algorithms used for data analysis ... describe data engineering and data preparation needs and solutions Application (3) • apply data engineering and data preparation concepts to design and implement a business intelligence and data analytics solution • implement a business intelligence dashboard including features of advanced data analytics					
3	Content • Business Intelligence and data Analytics (BIA): introduction and main concepts • BIA project management • Classical and modern BIA technical architecture • Data engineering for BIA solutions (data warehouse, data lake, data mesh, cloud solutions) • ETL (Extraction, Transforming, Loading) process • BIA systems and administration tools • Definition and implementation of KPIs for decision support • Introduction to machine learning for advanced data analytics • Machine learning algorithms and interfaces offered by common BIA vendors • State-of-the-art topics and market trends					
4	Teaching Methods Lecture and Practical					
5	Prerequisites Databases					
6	Methods of Assessment Graded Submitted Assignment and Presentation of Results (100%) including one milestone and one final presentation					

Version	Created by	Date (Date/Abbreviation)	Valid from
1.0	flm	14.02.2022/flm	01.03.2022

7	Application of Module Elective for International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Monika Frey-Luxemburger
9	Reading List • Chen, Hsinchun; Chiang, Roger H. L.; Storey, Veda C. (2012): Business intelligence and analytics. From big data to big impact. In: Management information systems : mis quarterly 36 (4), S. 1165–1188. • Davenport, Thomas H. (2014): Big data @ work. Dispelling the myths, uncovering the opportunities. Boston Massachusetts: Harvard Business Review Press. • Imhoff, C.; Gallemmo, N.; Geiger, J. G.: Mastering Data Warehouse Design – Relational and Dimensional Techniques. New York 2003. • Inmon, W. H.: Building the Data Warehouse. 4. Auflage, Indianapolis 2005. • Inmon, William H.; Linstedt, Dan (2015): Data architecture. A primer for the data scientist : big data, data warehouse and data vault. Amsterdam: Elsevier. • Kemper, H.-G.; Mehanna, W.; Unger, C.: Business Intelligence: Grundlagen und praktische Anwendungen – Eine Einführung in die IT-basierte Managementunterstützung. 2. ergänzte Auflage, Wiesbaden 2006. • Kimball, Ralph; Ross, Margy (2013): The Data Warehouse Toolkit. The Definitive Guide to Dimensional Modeling. 3. Aufl. s.l.: Wiley. • Kimball, R.; Caserta, J.: The data warehouse ETL toolkit – Practical techniques for extracting, cleaning, conforming, and delivering data. Indianapolis 2004. • Moss, Larissa Terpeluk (2013): Extreme scoping. An agile approach to enterprise data warehousing and business intelligence. S.l.: Technics Publications. • Thomsen, E.: OLAP Solutions – Building Multidimensional Information Systems. 2. Auflage, New York u.a. 2002. • Ng, Raymond Tak-yan (2013): Perspectives on business intelligence. San Rafael, Calif. (1537 Fourth Street, San Rafael, CA 94901 USA): Morgan & Claypool (Synthesis lectures on data management, # [34]). • Provost, Foster; Fawcett, Tom (2013): Data science for business. [what you need to know about data mining and data-analytic thinking]. 1. ed., 2. release. Beijing: O'Reilly. • Williams, Steve (2016): Business intelligence strategy and big data analytics. A general management perspective. Amsterdam, Boston, Heidelberg: Morgan Kaufmann.

Version	Created by	Date (Date/Abbriviation)	Valid from
1.0	flm	14.02.2022/flm	01.03.2022

Natural Language Processing and Text Mining

Module code	Workload	Credits/CP	Semester	Frequency of module		Duration
	180 h	6	1	Winter or Summer (available on request)		1 Semester
1	Module		Teaching Language	Contact hours	Self-study	Class size
	Natural Language Processing and Text Mining		English	45 h	135 h	30
2	Learning outcomes After passing this module successfully, students are able to ... Knowledge (1) - identify the component parts of natural language processing and text mining (NLP/TM) systems. - identify the application areas of NLP/TM systems. - identify a range of software tools for NLP/TM. Comprehension (2) - distinguish between text mining and natural language processing. - understand how such systems function. Application (3) - apply NLP and TM in business problems/situations, including the following. - Automation. - Interactive querying with databases. - Scanning social media for commentary. - Health monitoring. - Education. Analysis (4) - Deconstruct NLP/TM tools into their component parts. - Analyse situations in which an NLP/TM solutions would be appropriate. Synthesis (5) - develop applications based on NLP/TM tools to solve interaction problems in business situations. Evaluation (6) - test and evaluate critically the range of NLP/TM tools for implementing solutions.					
3	Individual component content Introduction to NLP/TM - Definitions, applications and history of NLP/TM - Basic ideas of computational linguistics. - Structure of NLP/TM systems. The Elements of NLP - Phonemes, Morphemes and Lexemes, Syntax and Context.					

	● Chomsky's categorisation of grammars. ● Structures for representing meaning and context. Constructing NLP/TM systems: Detailed PR Techniques. ● Introducing Pattern recognition and Machine Learning techniques for NLP. ● Categorising lexemes and meanings. ● Convolutional neural networks Constructing NLP/TM systems: The Text-Processing Pipeline ● From Computational Linguistic Models to Pipelines. ● Tokenization, Lemmatization, Tagging, Parsing, Entity Recognition ● Examination of pipelines in NLP/TM tools. Tools for NLP ● Python (spaCy, TensorFlow, Gensim) ● Text Mining with R. Applications of NLP/TM ● Database queries (content-based retrieval in documents and images). ● Chatbots. ● Data scraping (scanning social media for trending topics, monitoring misinformation, etc). ● Ethics and regulatory compliance.
4	Teaching methods Lectures, exercises and practices, presentations (Delivered Online). Practical applications will be discussed for training purposes. The classes will consist of discussions of basic theory, followed by examples based on Python coding samples and real world applications. Much of the learning will be derived from the assignment, which will be issued early in the semester and weekly classes will develop the knowledge required to complete the practical part of the assignment.
5	Prerequisites ● Applied Mathematics ● Introduction in Programming.
6	Methods of assessment Final written exam, presentation, written term paper
7	Applicability of module International Business Information Systems B.Sc.
8	Person responsible for module/ lecturer Prof. Dr. Leonard Noriega
9	Literature "Natural Language Processing and Computational Linguistics", B. Srinivasa-Desikan, Packt Publishing, 2018. "Natural Language Processing with Python and SpaCy: a practical introduction", Y. Vasiliev, No Starch Press, 2020. "Text Mining with Machine Learning: Principles and Techniques", J. Zizka, F. Darena, A. Svoboda, CRC Press, 2020. Appropriate journal articles and learned monographs according to the specific subject matter.

Signal Processing for Statistics and Data Science

Module code	Workload	Credits/CP	Semester	Frequency of module	Duration
	180 h	6	1	Winter Semester	Semester
1	Module	Teaching Language	Contact hours	Self-study	Class size
	An Introduction to Data Science	English	45 h	135 h	30
2	Learning outcomes After passing this module successfully, students are able to ... Knowledge (1) - list the data types used in data science - show awareness of the basic statistical approaches. Comprehension (2) - exemplify the relationship between data and digital signals. - demonstrate the strength and weaknesses of different algorithms and approaches. Application (3) - explain and use appropriate descriptive statistics for different data types. - deploy appropriate methodologies for the analysis of data. Analysis (4) - differentiate the different approaches to data science in contemporary use. - analyse problems in order to know which machine learning techniques to deploy. Synthesis (5) - make predictions using signal analysis. - develop applications based on machine learning techniques. Evaluation (6) - test and evaluate machine learning algorithms and applications.				
3	Individual component content An Introduction to data, signals and algorithms. - Data types - Algorithmic approaches. - Signal and sampling theory. Basic Signal Processing - Frequency domain transformations. - Operations in the frequency domain. Machine Learning - Unsupervised learning - distance measures and clustering. - Supervised learning - neural network learning. - Supervised learning - support vector machines.				

	Search Techniques • Standard search approaches. • Genetic Algorithms.
4	Teaching methods Lectures, exercises and practices, presentations
5	Prerequisites • Basic mathematics/statistics course. • Programming course.
6	Methods of assessment Final written exam, presentation, written term paper
7	Applicability of module Wherever applicable.
8	Person responsible for module/ lecturer Prof. Dr. Leonard Noriega
9	Literature • “Neural Networks and Learning Machines”, Simon Haykin, Pearson, 2008. • “Machine Learning”, Tom Mitchell, McGraw-Hill, 1997. • “Pattern Recognition: Statistical, Structural and Neural Approaches”, Robert Schalkoff, Wiley 1992. • “Digital Image Processing”, Rafael Gonzalez and Richard Woods, Addison Wesley, 2017. • Relevant articles from academic journals.

Business Strategy & Start-Up Development Project					
Module Code		Workload 180 hrs.	Credits 6	Semester 4. / 5. / 6. / 7.	Frequency Summer Semester (upon request)
Duration 1 Semester					
1	Module	Language	Contact hours	Self-study	Class Size
	Business Strategy Project	English	45 hrs.	135 hrs.	15
2	Learning Outcomes/Competencies The teaching event has less the character of a lecture than of a workshop. Students are given the opportunity to identify a given problem that requires a solution. Analyze the requirement, market, define relevant goals, tasks and work plans in teamwork and division of labor on a real-life use case in order to develop a solution or even implement the planning and prepare the founding and operation of a company. The use case may be conducted in a single semester or also extended over several semesters. Depending on the case setting, some of the following skills may be practiced. Knowledge (1) - Understand the given problem, identify the need for a solution, recognize and prioritize the necessary work. - Coordinate division of labor among team members and define goals, tasks and work plans. - Work result-oriented in the project, deliver usable results in time. Monitor project progress. Comprehension (2) - Analyze the state of development in the start-up phase of a company - Think across groups, i.e. in a networked manner, and thereby exchange insights that are useful for a solution - Report progress and interim results to the other groups. Collaborate across groups. Application (3) - Develop a business model and a revenue model for a given case or a company - Design offers, develop a product and a pricing strategy - Design and implement graphic design for the company's website and promotional materials - Select, plan, organize, implement, and evaluate appropriate sales measures and promotional activities. - Select, prepare and configure appropriate marketing channels and sales channels. Analysis (4) - Analyze market, identify market gap and analyze competitors - Identify target groups in the market, analyze their needs and quantify the demand potential. - Identify and evaluate appropriate sources of capital to finance a business start-up. Convince investors. - Critically examine a business idea for possible weaknesses and consider various szenarios. Synthesis (5) - Define core competencies of a start-up and identify potential business partners ("make or buy") - Write specifications and use cases, e.g. for the programming of an e-commerce website - Program frontend and backend of an e-business website - Prepare a financial and liquidity plan, develop a business plan and a financing concept - Prepare and conduct financial accounting and tax return. Select appropriate tools for operational management. - Perform proof of concept, evaluate different financing options and realize suitable ones				

	Learning Outcomes/Competencies Evaluation (6) • Evaluate business ideas. Perform profitability calculation and evaluate the revenue model. • Weigh up alternative positioning, growth and competitive strategies. Evaluate them for feasibility. • Evaluate possible legal forms for a business formation. • Draft articles of association, discuss with board members and subject to legal review.
3	Content a) In the Business Project module, students divide into working groups that take on different tasks similar to the departments of a company. A possible use case and corresponding contents of the project can be to formulate a business idea and prepare the foundation of a company or even to implement it practically and establish the company on the market. Therefore, there are many, possible work topics, but not all participants will work on them in the same depth. As a rule, the course consists of the following contents: • Introduction to the didactic intention of the course and to the evaluation criteria of the performance certificates to be provided. • Introduction to the problem and the initial situation of a business case. • Students divide themselves into individual working groups accordingly to the specific case, their respective knowledge, strengths and inclinations. • Similar to the departments of a company, the individual groups are each dedicated to a complex of topics, such as market research, business plan, organization, law, financing, controlling, marketing, sales, IT and programming of the company's website or e-shop. • Each group of students group defines its goals, tasks, measures, work packages and schedules. • Step by step, the foundation and operation of a company can be prepared. Starting with strategic planning, through a proof of concept, to clarification of financing and preparation of a foundation and (if desired and supported by students) the operational running of a company. • At the end of the semester, students reflect on the progress of the project and evaluate the performance of the respective working groups accordingly to the following criteria: Achievement of objectives, quality of task fulfillment, effectiveness of measures, quality of cooperation among the groups, results-oriented working methods, and operational usability of the work results.
4	Teaching Methods a) Seminar
5	Prerequisites Digital Marketing and e-Commerce Project Management Internship Semester (ideally, recommended)

6	Methods of Assessment a) Examination by submitted assignment (1A Practical work) (6 ECTS) The evaluation of the presented performance records of individual groups (each consisting of three persons) is done 50% by the students among themselves and 50% by the lecturer. The overall grade is made up of two parts (an arithmetic mean calculated of equally weighted two parts).
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person responsible for Module/Lecturer Prof. Dr. Pavel Rawe
9	Literature a) Students may conduct primary market and field research or develop an original solution to the case study. b) Robertson, Edward (2019): Strategic Planning - Process, Templates and effective Implementation c) Schwenker, Burkhard (2013): Scenario-based Strategic Planning – Developing Strategies in an uncertain World (Roland Berger School for Strategy and Economics) d) Schürly, Andreas (2020): Real Time Strategy – When Strategic Foresight meets Artificial Intelligence e) Nagle, Thomas (2017): Strategy and Tactics of Pricing – A Guide to Growing More Profitably f) Marr, Bernhard (2021): Data Strategy – How to profit from a World of Big Data, Analytics and Artificial Intelligence g) Porter, Michael (2004): Competitive Strategy – Techniques for Analyzing Industries and Competitors h) Garette, Bernard (2018): Cracked it! – How to solve big problems and sell solutions like top strategy consultants i) Safarova, Kristina (2021): Strategy Insights – Learn the skills used by the leading management consulting forms, such as McKinsey, BCG, et al. j) Plümer, Thomas (2016): Existenzgründung Schritt für Schritt – Mit 2 ausführlichen Business Plänen k) Kleindienst, Bernd (2017): Performance Measurement und Management – Gestaltung und Einführung von Kennzahlen- und Steuerungssystemen l) Nagl, Anna (2020): Der Businessplan – Geschäftspläne professionell erstellen – Mit Checklisten und Fallbeispielen m) Wiraeus, David (2018): Agile Strategy Management in the Digital Age – How dynamic Balanced Scorecards transform decision making, speed and effectiveness n) Helmer, Hamilton (2016): 7 Powers – The Foundations of Business Strategy o) Dib, Allan (2018): 1-Page Marketing Plan – Get new Customers, make more Money and stand out from the Crowd p) Weinhold, Stefan (2020): Das große Projektmanagement-Handbuch – traditionales, hybrides und agiles Projektmanagement – Grundlage, Checklisten, Arbeitshilfen für den Projekterfolg q) Harvard Business Review (2018): Harvard Business Review Entrepreneur’s Handbook – Everything you need to launch and grow your new business

Image Processing and Computer Vision							
Module code		Workload	Credits/CP	Semester	Frequency of module		Duration
		180 h	6	1	Winter or Summer (available on request)		1 Semester
1	Module			Teaching Language	Contact hours	Self-study	Class size
	Image Processing and Computer Vision			English	45 h	135 h	30
2	Learning outcomes After passing this module successfully, students are able to ... Knowledge (1) ● identify the component parts of image processing and computer vision (IP/CV) systems. ● identify the application areas of IP/CV systems. ● identify a range of software tools for IP/CV. Comprehension (2) ● understand the construction of IP/CV systems. ● understand how such systems function through the construction of pipeline processes. Application (3) ● apply IP/CV in business problems/situations, including the following. – Automation of business processes. – Content-based querying in multimedia databases. – Security applications (biometrics, facial recognition, behavior recognition, gesture recognition, etc). – Emotion monitoring. – Health monitoring. Analysis (4) ● Deconstruct IP/CV tools into their component parts. ● Analyse situations in which an IP/CV solutions would be appropriate. Synthesis (5) ● develop applications based on IP/CV tools for automation problems in business situations. Evaluation (6) ● test and evaluate critically the range of NLP/TM tools for implementing solutions (SCILab, Matlab, Python). .						
3	Individual component content An Introduction to Image Processing and Computer Vision ● The basic elements of low level IP/CV systems (raster scanning, spatial filtering). ● History and development of IP/CV systems. ● Construction of IP/CV systems - processing pipelines.						

	Representing image content. • Low-level vision - pixels, edges, segmentation. • Object recognition - shapes, colours, textures. • Representing spatial relationships. • Image Compression Video Analysis and Compression. • Analysing moving pictures. • Extending static image compression to moving images. Computer Vision and Artificial Intelligence • The relationship between Computer Vision, Machine Learning and Pattern Recognition. • Introducing Pattern recognition and Machine Learning techniques for Computer Vision. . • Convolutional neural networks in Computer Vision tasks. Constructing Computer Vision Systems: Tools and Building Blocks • Tools for constructing CV systems (SCILab, Matlab, Python+OpenCV). . • Constructing pipelines for IP/CV systems. Applications of IP/CV • Database queries (content-based of images and video). • Gesture and Facial Recognition. • Application areas: ○ Health ○ Education ○ Industry and Engineering. ○ Business and Commerce. • Ethics and regulatory compliance.
4	Teaching methods Lectures, exercises and practices, presentations(Delivered Online). Practical applications will be discussed for training purposes. The classes will consist of discussions of basic theory, followed by examples based on Python coding samples and real world applications. Much of the learning will be derived from the assignment, which will be issued early in the semester and weekly classes will develop the knowledge required to complete the practical part of the assignment.
5	Prerequisites None
6	Methods of assessment Final written exam, presentation, written term paper
7	Applicability of module Wherever applicable.
8	Person responsible for module/ lecturer Prof. Dr. Leonardo Noriega
9	Literature

- | | |
|--|---|
| | <ul style="list-style-type: none">● “Digital Image Processing”, R.C. Gonzalez and R.E. Woods, Pearson 2018.● “Image Processing, Analysis and Vision”, M. Sonka, V. Hladic and R. Boyle, 2015, .● “Mastering OpenCV 4 with Python : A Practical Guide Covering Topics from Image Processing, Augmented Reality to Deep Learning with OpenCV 4 and Python 3. 7”, A. Fernandez-Villan, Packt Publishing Ltd, 2019.● “Digital Image Processing with SCILab”, R.M. Thanki and A.M. Kothari, Springer International, 2019.● Journal Articles and Monographs for specific topic areas. |
|--|---|

Applied Mathematics					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	1	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Applied Mathematics	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students are able to ... Knowledge (1) ... name mathematical software. ... list subdomains within mathematics. ... name mathematical symbols. Comprehension (2) ... explain mathematical techniques. ... describe the interrelation between mathematical concepts. ... interpret mathematical expressions. Application (3) ... model mathematical problems with mathematical software. ... demonstrate mathematical concepts with mathematical software. ... determine appropriate approaches for solving mathematical problems. ... solve problems with mathematical concepts. ... use mathematical software for a given problem.				
3	Individual Component Content a) - Propositional calculus - Set theory - Sequences and series - Interest calculation - Linear algebra - Multivariable calculus - Mathematical optimization - Combinatorics - Probability calculus - Software for mathematics				
4	Teaching Methods a) Lecture / Practical				

5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (80%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung) ¹ a) Graded Assessment 1sbK (20%) (Written Exam) ¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Holger Ziekow (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Analysis for Computer Scientists : Foundations, Methods, and Algorithms,von Oberguggenberger, Michael, 1953- (GND-ID) ,und Ostermann, Alexander 1960- (GND-ID), VerfasserIn ,Veröffentlicht: Cham Springer, 2018,Auflage: 2nd ed. 2018 Comprehensive Mathematics for Computer Scientists 1 : Sets and Numbers, Graphs and Algebra, Logic and Machines, Linear Geometry (Second Edition),von Mazzola, Guerino 1947- (GND-ID) Milmeister, Gérard Weissmann, Jody ,Veröffentlicht: Berlin, Heidelberg Springer Berlin Heidelberg, 2006 Introduction to probability,von Blitzstein, Joseph K. ,und Hwang, Jessica ,Veröffentlicht: Boca Raton, FL CRC Press, [2015],Auflage: Online-Ausg.

¹ In the case of failure of one component of a multiple component assessment, all module component assessments must be repeated

Business Management					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	1	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Business Management	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... understand the basic concepts of Business Management. ... are able to describe business models, value chains and revenue cycles methods. Comprehension (2) ... are able to describe mega trends and resulting change of business models. ... are able to describe organizations of companies. Application (3) ... work on case studies of businesses (innovations also changes due to mega trends) Analysis (4) ... develop an eBusiness idea and the business model and analyze the competition.				
3	Individual Component Content a) Content - Mega Trends and how they shape Businesses today - Elements of an economy - Business organizations and their environments - Value Chain Models and Revenue Cycles - Business Models, Business Life Cycles and Change Management - Problem solving and management techniques in Business - Mergers and Aquisitions - Case Studies on the Mega Trends and Businesses - Development of an ebusiness idea for a startup - SWOT Analysis of the ebusiness idea - Development of a Business Plan				
4	Teaching Methods a) Lecture / Practical				

5	Prerequisites None
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (6 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Gabriele Hecker (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Dockery, Doug (2018): Modern Business Management. Creating a Built-to-Change Organization. Unter Mitarbeit von Laureen Knudsen. Berkeley, CA: Apress) Martínez-López, Francisco J. (2014): Handbook of Strategic e-Business Management. Berlin, Heidelberg: Springer). Tesch, Jan Fritz 1986 (2019): Business Model Innovation in the Era of the Internet of Things. Studies on the Aspects of Evaluation, Decision Making and Tooling. Cham: Springer International Publishing). Glauner, Friedrich (2016): Future Viability, Business Models, and Values. Strategy, Business Management and Economy in Disruptive Markets. Cham: Springer). Urbach, Nils; Röglinger, Maximilian (2019): Digitalization Cases. How Organizations Rethink Their Business for the Digital Age. Cham: Springer International Publishing).

International Business Information Systems B.Sc. (IBS) - SPO-Version 10 - Published: 22.01.2020

Professional Skills & Career Development							
Module Code		Workload	Credits	Semester	Frequency of Module	Duration	
		90 hrs.	3	1	Each semester	1 Semester	
1	Module Components		Teaching Language		Contact Hours	Self Study	Class Size
	a) Professional Skills & Career Development		a) Deutsch		a) 22,5 hrs.	a) 67,5 hrs.	a) 30
2	Learning Outcomes						
	After successful participation in the module the students ...						
	Knowledge (1)						
	... know about requirements for an effective cooperation in international and German teams.						
	... can describe methods on how to design and do presentations for specific target groups.						
	... are able to describe learning types and concepts and know about methods for identifying goals.						
	Application (3)						
	... are able to create their own personal learning concept for the bachelor program.						
	... are able to define their goals and then create a personal development plan.						
	... are able to plan and expand their professional network.						
	... are able to work out a professional presentation on a specific topic.						
	Analysis (4)						
	... are able to make an analysis on what the required milestones for a specific career goal are.						
3	Individual Component Content						
	a) 1. Development of personal learning strategies for studying Business information systems at Furtwangen University						
	- Welcome to a German University of Applied Sciences and what you need to know about the framework you are studying in						
	- Intercultural aspects						
	- What learning type are you?						
	- Time management						
	- Learning Methods: speed reading, ABC Lists, mindmaps, study apps						
	2. Personal career development plan						
	- Students will create their personal career development plan in the first semester and do updates continuously during their studies. The content of the development plan will also be used for the preparation of their apprenticeship application.						
	- The content of the development plan consists of personal goals, personal values, networking, strengths and weaknesses, social media strategy as well as planning of the career path.						

	3. Presentation techniques – how to do a professional presentation 4. Individual coaching sessions
4	Teaching Methods a) Seminar
5	Prerequisites None
6	Methods of Assessment a) Graded Assessment 1A (Practical Work) (3 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Gabriele Hecker (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Bolles, Nelson Richard: „What Color Is Your Parachute? A Practical Manual for Job-Hunters & Career-Changers“, Ten Speed Press, 2017 Chambers, Phil: “Brilliant speed reading : whatever you need to read, however you want to read it - twice as quickly“, Harlow, U.K. : Pearson, 2013 Covey, Stephen, PHD: „The seven habits of highly effective people: powerful lessons in personal change“, Mango Media, 2016, eBook HFU Library Flippo, Hyde: “When in Germany, do as the Germans do: the clued-in guide to German life, language, and culture“, Chicago ; London [u.a.] : McGraw-Hill, 2002 Lord, Richard: “CultureShock!, Germany : a survival guide to customs and etiquette“, Tarrytown, NY : Marshall Cavendish Editions, 2008, eBook HFU Library Timm, Paul R.: “Successful self-management: a psychologically sound approach to personal effectiveness“, Menlo Park, Calif.: Crisp Publications, c1993, eBook HFU Library Weissman, Jerry: “Don't make them think: creating the best flow for the elements of any great presentation” - Cover title - "An e-burst of inspiration from the best books.", Saddle River, N.J. : Pearson Education publishing as FTPress Delivers, c2010

Programming Fundamentals					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	1	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Programming Fundamentals	a) English	a) 45 hrs.	a) 75 hrs.	a) 30
	b) Programming Fundamentals Practicals	b) English	b) 22,5 hrs.	b) 37,5 hrs.	b) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... outline the concepts of imperative and structured programming. Comprehension (2) ... give original examples of the role of variables, functions and control structures. ... explain the difference between a compiler and an interpreter. Application (3) ... write computer programs of moderate complexity.				
3	Individual Component Content a) - Basic concepts of imperative programming - Variables - Subroutines - Control flow: choice, loops - Integrated Development Environments - Programming in Python 3 b) - Writing computer programs of moderate complexity				
4	Teaching Methods a) Lecture b) Practical				
5	Prerequisites Keine Eingabe vorhanden				
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (4 LP) b) Non Graded Assessment 1sbA (Practical Work) (2 LP)				

7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Ph.D. Andreas Hess (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Learn to program with Python 3, Irv Kalb, Apress, 2018 Beginning Python games development with PyGame (2nd ed.), Harrison Kinsley & Will McGugan, Apress, 2015 b) Python 3.7.3 documentation, The Python Software Foundation, online at https://docs.python.org/3/index.html

Scientific Working/ Academic Writing					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	90 hrs.	3	1	Only summer semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Scientific Working/Academic Writing	a) English	a) 22,5 hrs.	a) 67,5 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... list the common abbreviations and terms of a scientific apparatus ... name the necessary formal and content-related components of a scientific work ... recognise the boundary between plagiarism and correct quotation Comprehension (2) ... Describe the procedures for determining objectives ... understand the correct order of the steps of writing a scientific paper ... understand the formal rules for writing a scientific paper Application (3) ... find scientific, relevant questions ... adequately use the relevant research tools for your subject ... create a structure for a scientific work ... illustrate a lecture appropriately with slides ... prepare a scientific paper Analysis (4) ... Evaluate sources according to their relevance to the research question of a scientific paper				
3	Individual Component Content a) a) - Strategies of scientific writing - scientific research techniques - citation technique - Questions of the structure and structure of a scientific work - Presentation techniques and strategies of foil design - plagiarisms				

	- Scientific style - Formatting of a scientific paper
4	Teaching Methods a) Seminar
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1A (Practical Work) (3 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Marianne Andres (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Rowley, Jennifer; Slack, Frances (2004) "Conducting a literature review", Management Research News, Vol. 27 Issue: 6, pp.31-39, https://doi.org/10.1108/01409170410784185 Biggam, John (2013) „Succeeding with your Master's dissertation“, 3. Auflage, McGraw Hill, ISBN 978-0-335-26448-3 Saunders, Mark; Lewis, Philip; Thornhill, Adrian (2015) „Research Methods for Business Students“ Financial Times Prent., ISBN 978-1292016627 Denscombe, Martyn (2017) „The Good Research Guide: For Small Scale Research Projects“, 6th edition, Open University Press, ISBN 978-0335226863 Yin, Robert K. (2014) „Case study research : design and methods“, 5th edition, ISBN 978-1 -4522-4256-9

Systems and Network Architectures					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	1	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Systems and Network Architectures	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students are able to... Knowledge (1) ... explain the relevant models for Systems Architectures and Network Architectures. ... describe the role of information and communication technology for modern organisations. ... differentiate the concepts of Systems Architectures and Network Architectures. ... give an overview on important types of Networking protocols and technologies. ... give an overview on important types of Operating Systems processes, activities and functions. Comprehension (2) ... explain the interdependencies between different networking protocols. ... understand process characteristics of Operating Systems. Application (3) ... compare and evaluate various implementation alternatives for computing. ... develop and compare design solutions for Computer Networks. ... select and apply appropriate Systems Architectures and Network Architectures to fulfil requirements of modern organisations.				
3	Individual Component Content a) Individual component content System Architectures - IT Architectures and Models - Computer Architectures and Components - Operating Systems Requirements and Processes (e.g. Memory Management, Concurrency and Synchronisation, Input / Output) Network Architectures - Network Architectures and Models - Implementation and Design von Computer Networks - Networking Protocols (e.g. IEEE 802 and TCP/IP Protocols) - Planning and Operation of IT Infrastructures				

4	Teaching Methods a) Lecture / Practical
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (80%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung) ¹ a) Graded Assessment 1sbA (20%) (Practical Work) ¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Ph.D. Andreas Hess (Module Responsible) Prof. Dr. Martin Knahl (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Andrew Tanenbaum and Todd Austin, Structured Computer Organization, Pearson, 2016 Andrew Tanenbaum, Modern Operating Systems, Pearson, 2016 James Kurose and Ross Keith, Computer Networking: A Top-Down Approach, Global Edition, Pearson, 2016

¹ This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Applied Statistics					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	2	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Applied Statistics	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students are able to ... Knowledge (1) ... understand the importance of R in Statistics ... label statistical methods ... list statistical methods Comprehension (2) ... deal with statistical results ... describe statistical methods ... understand the importance of Statistics Application (3) ... solve statistical results ... analyze statistical results ... apply R in Statistics ... develop different test scenarios ... illustrate statistical results				
3	Individual Component Content a) Individual component content Numerical descriptive methods: - Mean, Median, and Mode - Geometric mean - Range, Variance, Standard deviation - Coefficient of variation - Z-scores, Quartiles - Boxplot Probability Theory				

- Probability
- Combinatorics

Discrete and continuous probability distributions:

- Population mean, variance, and standard deviation
- Expected value
- Variance and standard deviation
- Binomial distribution
- Poisson distribution
- Normal distribution

Forecasting:

- The importance of forecasting
- Forecasting methods
- Time frame of forecasts
- Components of demand patterns
- The process to forecasting demand
- Correlation coefficient
- Time series
- Coefficient of determination
- Seasonal patterns
- Forecast errors
- MAD and MAPD

Sampling distributions:

- of the mean
- of proportions
- standard error
- Central Limit Theorem

Confidence Interval Estimation:

- for mean (Sigma known and unknown)
- for proportion
- determining sample size

Fundamentals of Hypothesis Testing

- methods
- T-Test for mean (sigma unknown)
- One-tail test
- Z Test
- Critical value approach and p-value approach

Two-Sample Test

- comparing of means of two independent populations
- comparing of means of related populations
- F-test

Analysis of Variance

- One way ANOVA
- MANOVA
- Two-way Analysis of variance

Chi-Square and Nonparametric Tests

- Chi-Square tests
- Wilcoxon test
- Kruskal-Wallis test
- McNemar test

Simple linear regression

- Regression models
- Regression equations
- Measures of Variation
- Residual Analysis
- Autocorrelation
- Inferences (t-Test, F-test)
- Estimation

	Multiple regression - Modelling - Coefficient of multiple determination - Adjusted r^2 - Residual analysis - Inferences Logistic regression
4	Teaching Methods a) Lecture / Practical
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (80%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung)¹ a) Graded Assessment 1sbK (20%) (Written Exam)¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Thomas Marx (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) • D. Levine, D. Stephan, Statistics and Analytics, Pearson, 3rd edition, 2015 • Leonard J., Kazmier, Business Statistics, Schaum's outlines, 4th edition, 2009 • Peter Bruce, Andrew Bruce, Statistics for Data Scientists, O. Reilly, 2018 • Statistical Analysis with R, J. Schmuller, Wiley, 2017

¹ This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Business Processes					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	2	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Business Processes	a) English	a) 45 hrs.	a) 75 hrs.	a) 30
	b) Business Processes Practicals	b) English	b) 22,5 hrs.	b) 37,5 hrs.	b) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... know important areas of Business Process Management (process life cycles, modelling and improving processes). Comprehension (2) ... understand Business Process Orientation in companies and the impact of the current mega trends on those processes. ... understand that managing Key Business Processes is an important aspect about helping to execute the business strategy. Application (3) ... are able to apply the methods from areas of Business Process Management (process life cycles, modelling and improving processes). ... are able to create flowcharts of Business Processes and define inputs, outputs and metrics. Analysis (4) ... are able to identify Business Process problems, analyze it and work out an improvement				
3	Individual Component Content a) - Megatrends that influence and change existing Business Processes - Processes, Workflows, BPM life cycle - Digital driven BP innovations - Modelling techniques: Swimlanes, EPC, BPM - As-is Analysis and To-be Design - What Business processes do you need for your Business Idea (from Business Administration) b) In working teams of 3-4 members, the students practice the modelling of Business Processes in order to deepen the understanding and knowledge of the lecture. MS Visio and ARIS will used for the modelling.				

4	Teaching Methods a) Lecture b) Practical / Lab
5	Prerequisites None
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (4 LP) b) Non Graded Assessment 1sbA (Practical Work) (2 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Gabriele Hecker (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Davis, Robert E.: "ARIS Design Platform: Getting Started with BPM", Springer, 2007, eBook in HFU Library Dumas, Marlon; La Rosa, Marcello; Mendling, Jan; Reijers, Hajo A.: "Fundamentals of Business Process Management", Berlin, Heidelberg : Springer, 2018, eBook in HFU Library Kumar, Akhil., "Business Process Management" 1st Edition, Publisher: Routledge; 1 edition (February 14, 2018) Panagacos, Theodore: "The Ultimate Guide to Business Process Management: Everything you need to know and how to apply it to your organization", CreateSpace Independent Publishing Platform (25. September 2012) b) see literature for lecture

Data Bases					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	2	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Databases	a) English	a) 45 hrs.	a) 75 hrs.	a) 40
	b) Database Practicals	b) English	b) 22,5 hrs.	b) 37,5 hrs.	b) 20
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... are able to write simple and advanced SQL Statements ... are able to describe database basics ... are able to read and understand database models (e.g. Entity Relationship Model (ERM)) Comprehension (2) ... are able to describe the use of database systems in context of business information systems ... are able to design SQL statements ... are able to explain the meaning and use of database systems ... understand basic database concepts like normalization, database views and indices and transaction management Application (3) ... are able to design a relational database ... are able to implement a relational database				
3	Individual Component Content a) Individual component content (a) Databases lecture - Overview: Database systems and their fields of application - Difference between file-based and database-driven data management - 3- Schema architecture according to ANSI/SPARC - Database design (conceptual, logical and physical) - Entity Relationshipship Model (ERM) - Normalization				

	- Relational algebra - Structured Query Language (SQL) - Views and indices - ACID and transaction management b) Databases practicals (Assignments) - Entity Relationship Modeling (ERM) - Normalization - Simple SQL Statements - Advanced SQL Statements
4	Teaching Methods a) Lecture b) Practical
5	Prerequisites none
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (4 LP) b) Non Graded Assessment 1sbA (Practical Work) (2 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Monika Frey-Luxemburger (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) • Connolly, Thomas M., Begg, Carolyn E (2003).: Database Solutions: A Step by Step Guide to Building Databases: A Step-by-Step Approach to Building Databases. Addison Wesley; Auflage: 2. A. • Garcia-Molina, Hector; Widom, Jennifer; Ullman, Jeffrey D. (2009): Database systems. The complete book. Upper Saddle River, NJ: Prentice Hall. • Harrington, Jan L. (2016): Relational Database Design and Implementation. 4th ed.: Elsevier Science.

Digital Marketing and eCommerce					
Module Code	Workload 90 hrs.	Credits 3	Semester 2	Frequency of Module Only winter semester	Duration 1 Semester
1	Module Components		Teaching Language	Contact Hours	Self Study
	a) Digital Marketing and eCommerce		a) English	a) 22,5 hrs.	a) 67,5 hrs.
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... List actual developments in consumer markets and highlight consequences for marketing management ... Name analytical methods to develop a corporate strategy ... Differentiate the effect and success of advertising and assign comprehensive marketing metrics ... Draft working models and their elements to develop marketing plans, marketing strategies and brand messages ... Draft various models of digital distribution channels ... List various tools and types of media for the classical and digital promotion mix Comprehension (2) ... Differentiate strategies for growth and competitiveness and explain on examples ... Explain the difference between Push- and Pull-Promotion in classical and digital marketing and illustrate by examples ... Explain and illustrate on an example the interrelation between brand positioning, brand promise, online reputation and brand value ... List and explain success factors of social media advertising on the example of Instagram Application (3) ... Design the layout of a sales conversion oriented website for a given product ... Analyse a given product, formulate a brand positioning and design an appropriate marketing mix ... Select appropriate advertising media and policies for a given application case by cost, effectiveness and efficiency Evaluation (6) ... Evaluate the performance of a website, interpret the marketing metrics and suggest an optimization				
3	Individual Component Content a) - Introduction (key terms, devel. of technologies, consumer behaviour & marketing) - Scope of functions in marketing and sales management - Corporate strategies (analytical foundations, basic orientations and types) - Types of digital business models and marketing strategies - Marketing strategy, marketing plan and marketing mix (models and components) - Market & consumer research (research project, plan, quality criteria, data sources) - Market potential, consumer typologies, consumer segments, personas and profiles - Brand management, brand strategy, brand positioning, brand message, ad formats - Pricing strategy, dynamic pricing, yield and revenue management - Digital pricing, allocation and inventory management (methods and instruments)				

	- Online reputation monitoring & management, customer satisfaction, brand value - Multi channel distribution and cross channel advertising (strategy and operations) - Advertising media mix for classical mass marketing and digital marketing - Customer journey analysis, lead generation, UX design, conversion optimisation - Marketing automation and customer relationship management (methods, tools) - Marketing controlling (key metrics in marketing, digital advertising and eCommerce)
4	Teaching Methods a) Lecture / Practical
5	Prerequisites None
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (3 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Pavel Rawe (Module Responsible) Prof. Dr. Pavel Rawe (Lecturer)

9 Reading List (Core Texts and Recommended Texts)

- a) Robert Palmatier, Shrihari Sridhar (2017): Marketing Strategy – Based on first Principles and Data Analytics
- Orville Walker, John Mullins (2013): Marketing Strategy – A decision-focused Approach
- Mark Jefery (2010): Data-Driven Marketing – The 15 Metrics everyone in Marketing should know
- Mike Grigsby (2018): Marketing Analytics – A practical guide to improving Consumer Insights using Data Techniques
- Chuck Hemann, Ken Burbary
- Chuck Hemann, Ken Burbary (2018): Digital Marketing Analytics – Making Sense of Consumer Data in a Digital World
- Philip Kotler, Kevin Lane Keller (2017): Marketing Management, 15th Edition
- Alexander Chernev (2019): Strategic Marketing Management – The Framework, 10th Edition
- Simon Kingsnorth (2019): Digital Marketing Strategy – An integrated Approach to Online Marketing
- Ira Kaufman, Chris Horton (2014): Digital Marketing – Integrating Strategy and Tactics with Values, a guidebook for executives and students
- Robert W. Bly (2018): The Digital Marketing Handbook – A step-by-step Guide to Creating Websites that sell
- Akkan Dib (2018): The 1-Page Marketing Plan – Get new Customers, make more Money and stand out from the Crowd
- Drayton Bird (2007): Commonsense Direct & Digital Marketing
- Oliver Busch (2015): Programmatic Advertising – The successful Transformation to automated, data-driven Marketing in real-time
- Russell W. Belk (2012): Qualitative Consumer and Marketing Research
- Paul M.W. Hackett, Jessica Schwarzenbach (2016): Consumer Psychology – A study guide to qualitative Market Research
- Mario Mazzocchi (2008): Statistics for Marketing and Consumer Research
- Frank R. Kardes, Paul M. Herr (2019): Research Methods in Consumer Psychology
- Leon G. Schiffman, Joseph L. Wisenblit (2018): Consumer Behaviour – What's new in Marketing, 12th Edition

Financial Accounting					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	90 hrs.	3	2	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Financial Accounting	a) English	a) 22,5 hrs.	a) 67,5 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students are able to ... Knowledge (1) ... understand book keeping and posting records ... explain the main instruments of financial reporting Comprehension (2) ... explain how business transactions are recorded with the technique of double entry book keeping Application (3) ... apply the different tools of cost accounting in order to provide the necessary information for economic decisions within a company Analysis (4) ... carry out a cost deviation analysis				
3	Individual Component Content a) Individual component content - Financial accounting fundamentals - Recording business transactions and their impact on the Balance sheet - Asset side of the Balance sheet - Equity and Liability side of the Balance sheet - Income Statement - Statement of changes in equity - Cash-flow statement - Inventory and Cost of Goods Sold				
4	Teaching Methods a) Lecture / Practical				
5	Prerequisites Keine Eingabe vorhanden				

6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (3 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Thomas Marx (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) • Atrill, P./McLaney, E., Financial accounting for decision makers, 9th edition, 2019 • Harrison, W./Horngren/William/Wendy/Temin, Financial Accounting, 11th edition, 2017 • Libby, R./Libby, P./Short, D., Financial accounting, 9th edition, 2016

Object-oriented Programming					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	2	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Object-oriented Programming	a) English	a) 67,5 hrs.	a) 112,5 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... Learn specific programming tasks ... Complete knowledge in object oriented programming Comprehension (2) ... Define software architectures ... Program graphical user interfaces ... Understand Object oriented design Application (3) ... Program with and without intergrated development environment (IDE) Analysis (4) ... Analyse software design ... Analyse programming solutions				
3	Individual Component Content a) Object oriented design - Discussion of concepts Graphical user interfaces - Widgets - Listener - Composition of user interfaces Serialisation - Streams - File I/O Communication over IP				

	Collections and Generics Multi-Threading Exceptions
4	Teaching Methods a) Lecture / Practical
5	Prerequisites Basic principles in programming
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (6 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Stefan Noll (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Head First Java, O'Reilly and Associates, Kathy Sierra et al., 2005 Head First Design Patterns, O'Reilly and Associates, Eric Freeman et al., 2014 Object-Oriented Software Engineering Using UML, Patterns, and Java, Prentice Hall, Bernd Bruegge et al, 2013

Applied Algorithms and Data Structures					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	3	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Applied Algorithms and Data Structures	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students ... Comprehension (2) ... understand the basic concepts of optimisation problems ... understand the concepts of algorithms and complexity ... understand the concepts of lists, maps, trees and graphs ... understand the tasks of business programming Application (3) ... program with a document-oriented database ... apply the Java library for lists, maps, trees and graphs ... query a document-oriented database				
3	Individual Component Content a) Object oriented data structures - Interfaces and inheritance - Lists - Data structures - Sorting - Business programming with lists - Complexity of algorithms - Maps with hash functions - Data structures - Business programming with maps - Trees - Document-oriented databases with MongoDB - Business programming with databases - Business programming with XML and DOM - Maps with trees - Database index structures - Graphs and Shortest Path Algorithms				

4	Teaching Methods a) Lecture / Practical
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (6 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Ulf Schreier (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Cormen, T. H., Leiserson, C.E., et al.: Introduction to Algorithms, MIT Press, 2009 Sedgewick, R., Wayne, K.: Introduction to Programming in Java, Sams Publishing 2017 MongoDB, www.mongodb.com , Stand: 28.2.2019 The Java Tutorials, Trail: Java API for XML Processing (JAXP), https://docs.oracle.com/javase/tutorial/jaxp/index.html , Stand: 28.2.2019

ERP-Systems					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	3	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) ERP-Systems	a) English	a) 45 hrs.	a) 135 hrs.	a) 50
2	Learning Outcomes After successful participation in the module the students are able to ... Knowledge (1) ... • Demonstrate important processes and modules of ERP systems ... • Describe the integration principles in ERP systems ... • Explain logistics and accounting integration Comprehension (2) ... • Understand the relationships between logistics and finance / accounting ... • Classify various logistic processes ... • Describe complex logistic processes ... • Describe financial processes Application (3) ... • Explain the project phases of the introduction of ERP systems ... • Describe the object model ... • Explain an organizational model ... • Train and practice an ERP system				
3	Individual Component Content a) Individual component content - Accounting basics, master data, transaction data - Logistic basics, master data, transaction data - Special transactions in Finance, Controlling, Materials Management, Production Planning, Sales & Distribution				
4	Teaching Methods a) Seminar				
5	Prerequisites Keine Eingabe vorhanden				

6	Methods of Assessment a) Graded Assessment 1K (80%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung) ¹ a) Graded Assessment 1sbA (20%) (Practical Work) ¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Thomas Marx (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) • Sales and Distribution in SAP ERP, M. Chudy, L. Castedo, Galileo Press, 3rd edition, 2018 • Logistics with SAP ERP, M. Murray, Galileo Press, 2nd edition, 2013 • Production Planning and Control with SAP, J.T.Dickersbach, G. Keller, K. Weihrauch, Galileo Press, 2nd edition, 2012 • SAP MM – Functionality and Technical Configuration, M. Murray, J. Akhtar, 4th edition, Galileo Press, 2016 • Sales and Distribution, R. Lopez, A. Mohabatra, SAP Press, 2nd edition, 2016 • Product Cost Controlling with SAP, J. Jordan, Galileo Press, 3rd edition, 2016

¹ This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Introduction to Computer Security					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	90 hrs.	3	3	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Computer Security	a) English	a) 22,5 hrs.	a) 67,5 hrs.	a) 50
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... describe the top application security risks. ... list current encryption algorithms. ... recite the goals of IT security Comprehension (2) ... explain threat models. ... explain the basic legal aspects of IT security. Application (3) ... apply cryptography in order to secure communication. ... defend a system against the most common attacks. ... show attack vectors. Analysis (4) ... analyse risks and threats.				
3	Individual Component Content a) - security goals - introduction to basic legal aspects - threat modelling - application security risks - SQL injections - analysis of network communication with Wireshark - firewalls - introduction to cryptography - hash functions - public key infrastructures - secure communication protocols (e.g. TLS) - penetration testing with Metasploit - hacking IoT devices				

4	Teaching Methods a) Seminar
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (3 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Ph.D. Andreas Hess (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Stallings, William and Brown, Lawrie: Computer security: principles and practice, Pearson 2012 Stallings, William: Cryptography and network security, Prentice Hall 2011 Tanenbaum, Andrew and Wetherall, David: Computer networks, 5th edition, Pearson, 2014 Schneier, Bruce: Applied Cryptography, Applied cryptography : protocols, algorithms, and source code in C, 20th edition, Wiley 2015 Schneier, Bruce: Secrets and Lies: Digital Security in a Networked World, Wiley 2014 Sinha, Sanjib: Beginning Ethical Hacking with Kali Linux, Apress 2018 Rahalkar, Sagar: Quick Start Guide to Penetration Testing, Apress 2019

Introduction to Computer Security					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	90 hrs.	3	3	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Computer Security	a) English	a) 22,5 hrs.	a) 67,5 hrs.	a) 50
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... describe the top application security risks. ... list current encryption algorithms. ... recite the goals of IT security Comprehension (2) ... explain threat models. ... explain the basic legal aspects of IT security. Application (3) ... apply cryptography in order to secure communication. ... defend a system against the most common attacks. ... show attack vectors. Analysis (4) ... analyse risks and threats.				
3	Individual Component Content a) - security goals - introduction to basic legal aspects - threat modelling - application security risks - SQL injections - analysis of network communication with Wireshark - firewalls - introduction to cryptography - hash functions - public key infrastructures - secure communication protocols (e.g. TLS) - penetration testing with Metasploit - hacking IoT devices				

4	Teaching Methods a) Seminar
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (3 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Ph.D. Andreas Hess (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Stallings, William and Brown, Lawrie: Computer security: principles and practice, Pearson 2012 Stallings, William: Cryptography and network security, Prentice Hall 2011 Tanenbaum, Andrew and Wetherall, David: Computer networks, 5th edition, Pearson, 2014 Schneier, Bruce: Applied Cryptography, Applied cryptography : protocols, algorithms, and source code in C, 20th edition, Wiley 2015 Schneier, Bruce: Secrets and Lies: Digital Security in a Networked World, Wiley 2014 Sinha, Sanjib: Beginning Ethical Hacking with Kali Linux, Apress 2018 Rahalkar, Sagar: Quick Start Guide to Penetration Testing, Apress 2019

Management Accounting and Controlling					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	3	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Management Accounting and Controlling	a) English	a) 45 hrs.	a) 135 hrs.	a) 50
2	Learning Outcomes After successful participation in the module the students are able to... Knowledge (1) ... • describe the main elements of a cost accounting system Comprehension (2) ... • apply the different tools of cost accounting in order to provide the necessary information for economic decisions within a company ... • explain how business transactions are recorded with the technique of double entry book keeping Application (3) ... • illustrate the common tools of cost accounting and their differences Analysis (4) ... • compare different cost accounting tools and assess their advantages and disadvantages ... • carry out a cost deviation analysis				
3	Individual Component Content				
4	Teaching Methods a) Lecture / Practical				
5	Prerequisites Keine Eingabe vorhanden				
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (6 LP)				
7	Applicability of Module International Business Information Systems B.Sc. (IBS)				
8	Person Responsible for Module Prof. Dr. Thomas Marx (Module Responsible)				

9	Reading List (Core Texts and Recommended Texts) a) • Atrill, P./McLaney, E., Management accounting for decision makers, 8th edition, 2015 • Bhimani, A./Horngren, Ch./Datar, S./Rajan, M.; Management and Cost Accounting, 6th edition, 2015 • Horngren, Ch. et al., Management and Cost Accounting, 7th edition, 2019 • Drury, C, Management Accounting for Business, 6th edition, 2016
---	--

Project Management					
Module Code	Workload 90 hrs.	Credits 3	Semester 3	Frequency of Module Each semester	Duration 1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Project Management	a) English	a) 22,5 hrs.	a) 67,5 hrs.	a) 50
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... can show the importance of project work for a company and the interaction between projects and the company ... explain the project initiation process group ... give an overview of the processes and knowledge areas of project management ... know important techniques of project planning ... know the psychological and economic foundations of leading project teams ... provide an overview of procedures for monitoring project progress Comprehension (2) ... explain the significance of risk management in projects and explain risk management techniques ... describe the cooperation with project stakeholders ... elaborate on the soft skills required for the management of projects ... explain the qualifications of a project manager and be able to understand the changes in this profession that are emerging as a result of digitisation ... present the basic principles of the agile project management method Scrum and map them into the knowledge canon of project management Application (3) ... initiate, plan and monitor the progress of a small (student) project using agile project management methods (Kanban boards, self-organized teams, ...)				
3	Individual Component Content a) The course is based on the body of knowledge developed by the Project Management Institute (PMI). Basics from all three fields of the PMI Talent Triangle (technical project management, leadership, business management and strategy) are covered, with a technical project management emphasis. The following topics are covered: - Project Management and skills of a project manager - Project life cycle (project management process groups, phases, agile and plan-driven project life cycles, Scrum as an example of an agile project management approach) - Interaction between a project and the enterprise - Working with stakeholders				

	- Project initiation (project goals, project charter, team building) - Project closure - Decision making in projects (models for decision making – rational agent, bounded rationality, behavioural economics, heuristics and biases, decisions under uncertainty, group decisions) - Cost/benefit analysis and scope planning - Estimation procedures, scheduling, cost planning and budgets - Metrics and KPIs for project management - Performance monitoring with completion percentages and the earned value method - Status reports - Change and issue management - Project quality management - Risk management in projects - Security and project work - Organizing a project team and strategies to improve productivity of the project staff - Self-organized teams - Basics of leadership and motivation - Virtual teams and intercultural cooperation - Effects of diversity on project work (in particular gender) The methods taught in the course are applicable to all types of projects. They are often explained using software engineering projects as an example.
4	Teaching Methods a) Seminar
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (50%) (Written Exam) (3 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung) ¹ a) Graded Assessment 1sbA (50%) (Practical Work) ¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Marianne Andres (Module Responsible)

9	Reading List (Core Texts and Recommended Texts) a) Project Management Institute, A Guide to the Project Management Body of Knowledge (Pmbok Guide), 6th edition, 2017 Kerzner, Project management, 12 ed., Wiley 2017 Mike Cohn, Succeeding with Agile. Addison-Wesley 2010 Louise Worsley, Stakeholder-Led Project Management, Business Expert Press, New York 2017 Peter F. Drucker, Management Challenges for the 21st Century, Butterworth-Heinemann, Burlington 2007 Erin Meyer, The Culture Map. Decoding how people think, lead, and get things done across cultures. Public Affairs, New York 2015 Daniel Kahneman, Thinking, Fast and Slow,, Farrar, Straus and Giroux, New York 2011 Further literature (including Internet sources and multimedia content) will be given by the lecturer
---	--

¹ This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Web-based Technologies					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	3	Only summer semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Web-based Technologies	a) English	a) 67,5 hrs.	a) 112,5 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... know relevant basics from perceptual and cognitive psychology like Gestalt Theory for dealing with user interfaces ... sketch possibilities for the technical implementation of web projects ... outline communication models and their application to user interface design. ... present the models for action processes. ... recognize the most important interaction concepts for the creation of user interfaces Comprehension (2) ... explain the structure of client/server Internet architecture ... construct information architecture and navigation of websites due to the needs of target groups ... judging the design of usable and user-friendly websites and applications ... select the relevant methods for evaluating usability Application (3) ... self-contained preparing and presenting of selected topics for the website conception ... implement dynamic websites using client side web-technologies like HTML, CSS, JavaScript ... Use mock-up tools for prototyping (mobile) websites Analysis (4) ... design and perform a usability test for the prototype and evaluate the results				
3	Individual Component Content a) a) User centered design, methods and principles - Mental models - Action processes and interaction models - Affordances and mappings Website Conception - Rough conception: Design Thinking, user needs, personas				

	- Fine conception: information and navigation architecture - Design concept: layout - responsive design, text, colors, assets, interactive elements Implementation technologies of websites - Prototyping - HTML 5 + CSS3, interactive elements with JavaScript - Content Management Systems CMS - Search engine optimization SEO Cognitive Science and perception psychology - Perception - Memory (ACT*-Model) and Experience - Communication and interaction concepts Usability Evaluation - analytical & empirical test methods b) Implementation of an own project - Introduction to Mockup Tools - Development and implementation of the rough, fine and design concept - Execution of usability tests in a usability laboratory
4	Teaching Methods a) Lecture / Practical
5	Prerequisites Programming Fundamentals
6	Methods of Assessment a) Graded Assessment 1K (70%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung) ¹ a) Graded Assessment 1sbA (30%) (Practical Work) ¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)

8	Person Responsible for Module Prof. Dr. Marianne Andres (Module Responsible) Prof. Dr. Oliver Tamine (Lecturer)
9	Reading List (Core Texts and Recommended Texts) a) John Duckett, Web Design with HTML, CSS, JavaScript and jQuery Set 1st Edition Steve Krug, Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability (3rd Edition) (Voices That Matter) 3rd Edition Nielsen, Jakob; Loranger, Hoa: Web Usability, Addison-Wesley 2006 Nielsen, Jakob; Budiu, Raluca: Mobile usability : für iPhone, iPad, Android und Kindle, 1. Aufl., mitp 2013 Norman, Donald A.: The design of everyday things, Rev. and expanded ed, Basic Books 2013 (E-Book) Shneiderman, Ben; Plaisant, Catherine: Designing the user interface : strategies for effective human-computerinteraction, 5. ed., Pearson/Addison-Wesley 2009 Anderson, John R., The Architecture of Cognition, Psychology Press 1995

¹ This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Cloud and Mobile Business Applications					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	5	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Cloud and Mobile Applications	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... differentiate between client and server applications ... describe microservices ... give an overview on important types of supply chain processes, activities and functions Comprehension (2) ... understand the architecture of mobile applications ... understand the architecture of cloud applications Application (3) ... develop a mobile application ... develop a cloud service				
3	Individual Component Content a) Individual component content - Introduction to cloud applications - Describe microservices - Development of microservices (e.g. REST) - Introduction to mobile applications - Development of a mobile App - Connecting the mobile App to the microservices				
4	Teaching Methods a) Lecture				
5	Prerequisites Keine Eingabe vorhanden				
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (6 LP)				

7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Oliver Tamine (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Erl, T., Puttini, R., Mahmood, Z.: Cloud Computing: Concepts, Technology & Architecture, Prentice Hall 2013 Reese, G.: Cloud Application Architectures: Building Applications and Infrastructure in the Cloud, O'Reilly 2009 Newman, S.: Building Microservices: Designing Fine-Grained Systems, O'Reilly 2015 Further and online references are given in course

Data Science and Machine Learning					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	5	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Data Science and Machine Learning in Business	a) English	a) 67,5 hrs.	a) 112,5 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students are able to ... Knowledge (1) ... outline concepts of data science. ... list data science methods and tools. Comprehension (2) ... use data science techniques to address business problems. ... describe the strength and weaknesses of data science methods. ... discuss the applicability of data science methods to business problems. Application (3) ... perform test to validate machine learning models. ... apply software tools to analyse and examine data. ... Illustrate solution options for data science approaches. ... Use science tools of analysis.				
3	Individual Component Content a) Individual component content - Concepts and tools of Data Science - The data science process - Application areas of data science and machine learning - Software for data science - Pre-processing - Data integration - Data cleaning - Exploratory data analysis - Data visualization - Descriptive statistics - Building machine learning models - Supervised and unsupervised Learning - Classification mechanisms - Regression models - Clustering mechanisms				

	- Evaluating machine learning models - Error measures and visualizations - Test strategies Evaluation from a business perspective
4	Teaching Methods a) Lecture / Practical
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1K (50%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung) ¹ a) Graded Assessment 1sbA (50%) (Practical Work) ¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Holger Ziekow (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Foster Provost, Tom Fawcett: Data Science for Business- What You Need to Know about Data Mining and Data-Analytic Thinking, O'Reilly Media, Inc., August 2013, ISBN-10: 1449361323 Cathy O'Neill: "Doing Data Science", published by O'Reilly Media, Inc., 2013, ISBN-10: 1449358659 Sandya Mannarswamy. Data Science: Learn the What, Where, and How of Data Science, published by Apress, 2015, ISBN10 1430261188

¹ This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Internet of Things for Business Solutions					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	6	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Internet of Things	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... describe IoT standards ... understand IoT protocols ... recognize Industrial IoT applications (e.g. SAP Leonardo) Comprehension (2) ... describe IoT processes ... design IoT interoperability Application (3) ... design IoT applications ... develop an IoT application				
3	Individual Component Content a) Individual component content - Introduction to Internet of Things - Standard and protocols - IoT platforms - Industrial applications - IoT Programming				
4	Teaching Methods a) Lecture / Practical				
5	Prerequisites Keine Eingabe vorhanden				
6	Methods of Assessment a) Graded Assessment 1K (Written Exam) (6 LP)				

7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Oliver Tamine (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Cirani, S., Ferrari, G., Piconi, M., Veltri, L.: Internet of Things: Architectures, Protocols and Standards, Wiley 2018. Lea, P.: Internet of Things for Architects: Architecting IoT solutions by implementing sensors, communication infrastructure, edge computing, analytics, and security, Packt Publishing 2018 Further and online references are given in course

Operations and Logistics					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	5	Only summer semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Operations and Logistics	a) English	a) 45 hrs.	a) 135 hrs.	a) 30
2	Learning Outcomes After successful participation in the module the students are able to ... Knowledge (1) ... understand process characteristics of operations and logistics systems. ... describe the role of information and communication technology for industrial networks. Comprehension (2) ... describe the potential of enabler technologies such as Auto-ID, IoT and cloud computing for operations and logistics processes. ... give an overview on important types of operations and logistics processes, activities and functions. ... understand the principles of modeling for operations and logistics processes using EPC and BPMN/DMN notation. Application (3) ... select and apply appropriate operations research methods ... model operations and logistics processes using EPCs and BPMN/DMN diagrammes. Analysis (4) ... analyse and evaluate problems in operations and logistics processes (as-is analysis). ... outline to-be processes for basic operations and logistics systems based on the identified requirements.				
3	Individual Component Content a) - Fundamentals of business operations and logistics - Examples from different industries, services and regions - Processes, systems and models in context with Industry 4.0 and CPS - Enabling technologies for logistics processes (incl. Auto ID, IoT, cloud solutions) - Operations and logistics strategy - Sourcing - Production (incl. concepts of additive manufacturing, mass customization) - Distribution and transportation - Return processes - Operations and logistics analytics - Network strategy including aspects of sustainability and resilience				
4	Teaching Methods a) Seminar				

5	Prerequisites Business management Business processes Basic principles in ERP systems
6	Methods of Assessment a) Graded Assessment 1K (70%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung) ¹ a) Graded Assessment 1sbA (30%) (Practical Work) ¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Guido Siestrup (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Ivanov, D. / Tsipoulaidis, A. / Schönberger, J. (2019) Global Supply Chain and Operations Management: A Decision-oriented Introduction to the Creation of Value. Springer. (eBook) ten Hompel, M. / Rehof, J. / Wolf, O. (2015) Cloud Computing for Logistics. Springer. (eBook) Zijm, H. / Klumpp, M. / Regattieri, A. / Heragu, S. (2019) Operations, Logistics and Supply Chain Management. Springer. (eBook) Further references are given in class.

¹ This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Software Engineering					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	5	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Software Engineering	a) English	a) 45 hrs.	a) 135 hrs.	a) 50
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... name the most important process models in software development with advantages and disadvantages ... describe the design methods of modern SW development ... describe the requirements process with quality criteria ... sketch test procedures Comprehension (2) ... describe quality assurance measures in SW development including test methods ... compare approaches and methods in different development processes ... recognize the relevance of a well-founded requirements analysis Application (3) ... create interaction diagrams ... analyze use cases and display them with UML including activity diagrams ... apply methods to collect requirements and systematically write them down ... create subject classes and design class models using selected design patterns ... create the SW design of a simpler SW application (e.g. web application with shop component) ... design a context diagram and the basic architecture				
3	Individual Component Content a) Contents Presentation and comparison of different models for software development processes - Waterfall model, V-Modell XT, iterative and agile methods - Important aspects are the customer and stakeholder relationship, teamwork, the risk minimization and quality management The requirement process - Role in the development process and procedure				

	- Quality Criteria and Problems of RE Process - Communicative aspects with survey methods - Define stakeholders, actors and context Methods and concepts of object-oriented analysis: - Business Use Cases, Scenarios, Use Case Diagrams - Activity diagrams (UML) with objects - Developing a business and design class model according to UML - Selected design patterns - Sequence diagrams on different levels quality assurance - Validation and verification - Design of tests on different levels
4	Teaching Methods a) Lecture / Practical
5	Prerequisites - Programming Fundamentals - Business Processes
6	Methods of Assessment a) Graded Assessment 1K (70%) (Written Exam) (6 LP insgesamt für alle Teilprüfungsleistung dieser Lehrveranstaltung)¹ a) Graded Assessment 1sbA (30%) (Practical Work)¹
7	Applicability of Module International Business Information Systems B.Sc. (IBS)

8	Person Responsible for Module Prof. Dr. Marianne Andres (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Craig Larman, Applying UML and Patterns, An Introduction to Object-Oriented Analysis and Design, Prentice Hall Suzanne Robertson, James Robertson, Mastering the Requirements Process, Addison-Wesley 1999 Ian Sommerville, Software Engineering, Pearson 2011 Eric Evans, Domain-Driven Design: Tackling Complexity in the Heart of Software, Addison-Wesley

¹

This graded assessment is only considered passed when all components of the assignment have received a minimum grade of "adequate", (4.0).

Business Information Systems (BIS) Project					
Module Code	Workload 180 hrs.	Credits 6	Semester 6	Frequency of Module Each semester	Duration 1 Semester
1	Module Components		Teaching Language	Contact Hours	Self Study
	a) BIS Project Task		a) English	a) 9 hrs.	a) 141 hrs.
	b) BIS Project Seminar		b) English	b) 11,25 hrs.	b) 18,75 hrs.
2	Learning Outcomes After successful participation in the module the students ... Knowledge (1) ... Present business processes for various organizations (institutions, companies) Comprehension (2) ... Report progress and interim results to the other groups ... Collaborate across groups, think in a networked way, exchange useful solutions/ideas ... Understand the problem, identify the need for solutions, identify and prioritise necessary work Application (3) ... Programme the frontend and backend of an e-business website (depending on the given case) ... Design attractive offers for a given demand and develop a pricing strategy ... Develop a business and revenue model and examine their feasibility ... Select, plan and implement suitable sales and advertising activities ... Work on the project in a result-oriented manner, deliver results that can be used just in time and monitor the progress of the project Analysis (4) ... Identify target groups in the market, illustrate their needs and quantify potential demand ... Analyse markets, identify market gaps and analyse competitors ... Critically examine a business idea in terms of possible weaknesses ... Develop solutions for a given problem Synthesis (5) ... Draw up a financial and liquidity plan, develop business plans and financing concepts ... Write specifications and use cases to programme an eCommerce website Evaluation (6) ... Process a proof of concept, evaluate various financing options and examine their feasibility ... Weigh up alternative growth, competitive and positioning strategies and evaluate their feasibility				
3	Individual Component Content a) Solutions for a given problem				

	Development of a specific application Innovative concepts Change and innovation management Process analysis Market analysis Development of business models and business plans Implementation of concepts b) Project management
4	Teaching Methods a) Project b) Seminar
5	Prerequisites Business economics Business processes and innovation management Project Management
6	Methods of Assessment a) Graded Assessment 1A (Practical Work) (5 LP) b) Non Graded Assessment 1sbPN (Presentation) (1 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Pavel Rawe (Module Responsible)

9	Reading List (Core Texts and Recommended Texts)
----------	--

- | | |
|----|---|
| a) | <p>Blank S., Dorf B. (2012): The Startup's owner's manual, 1. edit., John Wiley & Sons Inc.</p> <p>Evans D., Schmalensee R. (2016): Matchmakers – The news economics of multisided Platforms, 1. edit., Harvard Business School Press</p> <p>Teece D. (2009): Business Models, Business Strategy and Innovation, Article in Journal for Long Range Planning Nr. 43 (2010), p. 172-194</p> <p>Osterwalder A., Pigneur Y. (2010): Business Model Generation – A Handbook for Visionaries, Game Changers and Challengers, 1. edit. John Wiley & Sons Inc.</p> <p>Rogers D. (2016): The digital transformation playbook – rethink your business for the digital age, 1. edit., Columbia University Press</p> <p>Weitnauer W. (2019): Handbook Venture Capital, 1. edit., C.H. Beck Verlag</p> |
|----|---|

Thesis					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	540 hrs.	18	7	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Bachelor Thesis	a) English	a) 0 hrs.	a) 360 hrs.	a) 50
	b) Thesis Seminar	b) English	b) 0 hrs.	b) 180 hrs.	b) 50
2	Learning Outcomes				
	After successful participation in the module the students are able to ...				
	Knowledge (1) ... reflect the fundamentals of academic writing ... know structure and composition of scientific publications ... understand the relevant contents and methods learnt during the previous semesters				
	Comprehension (2) ... recognize the interdependencies between processes, structures, systems and institutions in context with the thesis topic				
	Application (3) ... apply acquired knowledge and qualifications to specific scientific topics in order to independently work on scientific problems in a sufficient depth and breadth ... select, explain and apply appropriate scientific methods on a specific problem description				
	Analysis (4) ... analyze a selected topic of higher complexity independently using scientific methods				
	Synthesis (5) ... contrast scientific theories with applied implementation results from practice				
	Evaluation (6) ... classify and evaluate the thesis results				
3	Individual Component Content				
	a) Familiarization with the thesis subject and setup of a work schedule. Experimental and/or theoretical work on a scientific subject. Documentation of the results by authoring the thesis. Presentation and discussion of the results.				

	b) Individual meetings with the supervisory team.
4	Teaching Methods a) b) Seminar
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1T (Thesis) (12 LP) b) Graded Assessment 1sbR (Review) (6 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Guido Siestrup (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Selection of literature depends on the topic.

Oral Exam					
Module Code	Workload	Credits	Semester	Frequency of Module	Duration
	180 hrs.	6	7	Each semester	1 Semester
1	Module Components	Teaching Language	Contact Hours	Self Study	Class Size
	a) Oral Exam	a) English	a) 0 hrs.	a) 180 hrs.	a) 50
2	Learning Outcomes The learning objectives refer to the preparation (self-study) of the oral exam. The oral exam shows the fulfilment of these learning objectives. After passing this module successfully, students are able to ... Knowledge (1) ... • describe relevant methods of scientific work Comprehension (2) ... • understand and classify expert knowledge of individual subject areas ... • represent a broad basic knowledge Application (3) ... • select, explain and apply scientific methods and findings for specific research questions Analysis (4) ... • analyze a selected topic of higher complexity independently using scientific methods				
3	Individual Component Content a) Two topics from different areas form the basis of the oral exam: Examination Area 1: Software and systems Examples: Data structures and algorithms, eBusiness, requirements engineering, testing and quality assurance, operating systems, computer architectures, computer networks, network architectures, network design, security management, internet technology, software design, software architectures, software engineering, programming languages, object orientation and object-oriented modeling, generative and model-based approaches, service-oriented architectures, web services, web applications, databases, data warehouse, business intelligence, application servers, markup languages, standard Examination Area 2: Business and management Examples:				

	Information and knowledge management, change management, customer relationship management, logistics and supply chain management, innovation management, project management, strategic planning, IAS and US GAAP, marketing strategies, market segmentation, controlling tools, controlling concepts, accounting systems, financial statements and financial statements analysis, balanced scorecard, integration concepts, business process design and management, eBusiness, risk management and controlling, cost accounting, financial services trends.
4	Teaching Methods a)
5	Prerequisites Keine Eingabe vorhanden
6	Methods of Assessment a) Graded Assessment 1M (Oral Exam) (6 LP)
7	Applicability of Module International Business Information Systems B.Sc. (IBS)
8	Person Responsible for Module Prof. Dr. Guido Siestrup (Module Responsible)
9	Reading List (Core Texts and Recommended Texts) a) Literature corresponds to the examination topics (examination area 1 + 2)